

Exp No:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# -----
# 1. CREATE DATASET
# -----

data = {
    "Date": [
        "01-01-2023", "02-01-2023", "03-01-2023", "04-01-2023",
        "05-01-2023", "06-01-2023", "07-01-2023", "08-01-2023",
        "09-01-2023", "10-01-2023", "11-01-2023", "12-01-2023",
        "13-01-2023", "14-01-2023", "15-01-2023", "16-01-2023"
    ],
    "Product": [
        "Product A", "Product B", "Product A", "Product C",
        "Product B", "Product A", "Product C", "Product B",
        "Product A", "Product C", "Product B", "Product A",
        "Product C", "Product B", "Product A", "Product C"
    ],
    "Sales": [
        200, 150, 220, 300, 180, 210, 320, 160,
        230, 310, 190, 240, 330, 170, 250, 340
    ],
    "Quantity": [
        4, 3, 5, 6, 4, 5, 7, 3,
        6, 7, 4, 6, 8, 3, 7, 8
    ],
    "Region": [
        "North", "South", "North", "East",
        "West", "North", "East", "South",
        "North", "East", "West", "North",
        "East", "South", "North", "East"
    ]
}

# Create DataFrame
df = pd.DataFrame(data)

# Convert Date column
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)

# -----
# 2. DISPLAY DATA
# -----
```

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print("DATASET:")
print(df.head())

print("\n")

# -----
# 3. CHECK MISSING VALUES
# -----

print("MISSING VALUES:")
print(df.isnull().sum())

print("\n")

# -----
# 4. STATISTICAL SUMMARY
# -----

print("STATISTICAL SUMMARY:")
print(df[["Sales", "Quantity"]].describe())

print("\n")

# -----
# 5. PRODUCT-WISE SALES AND QUANTITY
# -----

product_summary = df.groupby("Product")[["Sales", "Quantity"]].sum().reset_index()

print("PRODUCT-WISE SALES AND QUANTITY:")
print(product_summary)

print("\n")

# -----
# 6. LINE GRAPH - SALES OVER DATE
# -----

plt.figure(figsize=(10, 5))

plt.plot(
    df["Date"],
    df["Sales"],
    marker="o",
    color="blue",
    linewidth=2
)

plt.title("Sales Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")

```

```

plt.xticks(rotation=45)
plt.grid(True)

plt.tight_layout()
plt.show()

# -----
# 7. BAR GRAPH - PRODUCT-WISE SALES
# -----

plt.figure(figsize=(8, 5))

plt.bar(
    product_summary["Product"],
    product_summary["Sales"],
    color=["blue", "orange", "green"]
)

plt.title("Product-wise Total Sales")
plt.xlabel("Product")
plt.ylabel("Total Sales")

plt.tight_layout()
plt.show()

# -----
# 8. CORRELATION MATRIX
# -----

correlation = df[["Sales", "Quantity"]].corr()

print("CORRELATION MATRIX:")
print(correlation)

print("\n")

# -----
# 9. CORRELATION HEATMAP
# -----

plt.figure(figsize=(7, 5))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    linewidths=0.5,
    fmt=".2f"
)

plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()

```

DATASET:

	Date	Product	Sales	Quantity	Region
0	2023-01-01	Product A	200	4	North
1	2023-01-02	Product B	150	3	South
2	2023-01-03	Product A	220	5	North
3	2023-01-04	Product C	300	6	East
4	2023-01-05	Product B	180	4	West

MISSING VALUES:

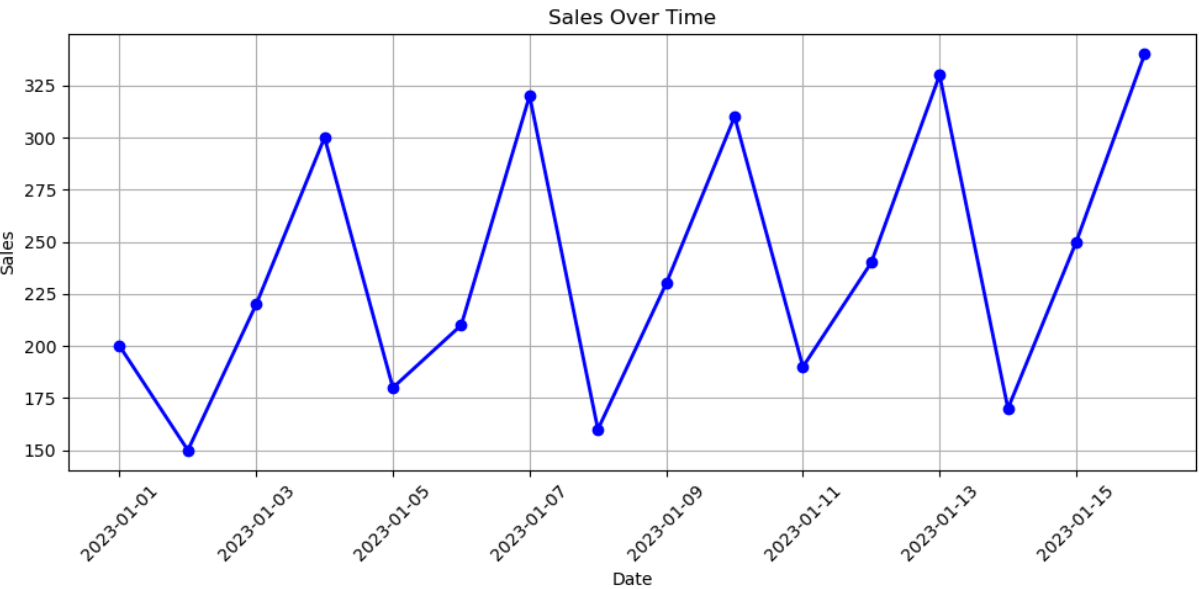
Date 0
Product 0
Sales 0
Quantity 0
Region 0
dtype: int64

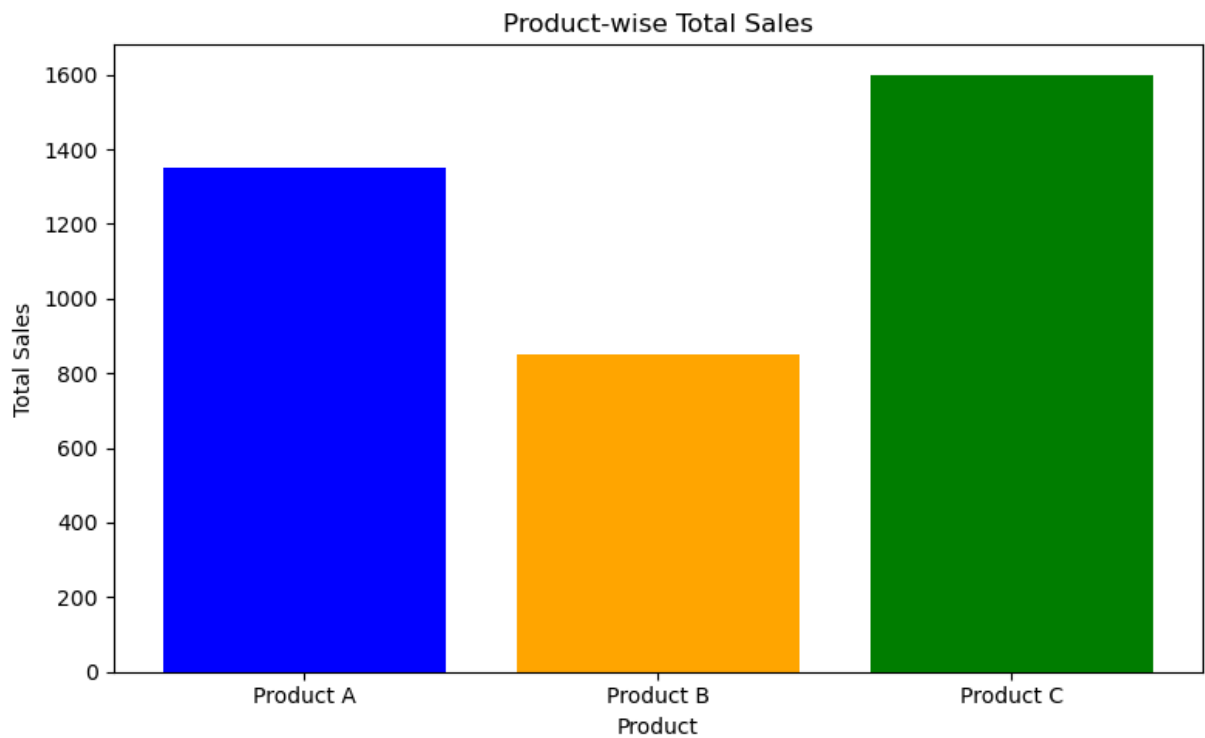
STATISTICAL SUMMARY:

	Sales	Quantity
count	16.000000	16.000000
mean	237.500000	5.375000
std	64.031242	1.746425
min	150.000000	3.000000
25%	187.500000	4.000000
50%	225.000000	5.500000
75%	302.500000	7.000000
max	340.000000	8.000000

PRODUCT-WISE SALES AND QUANTITY:

	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	17
2	Product C	1600	36





CORRELATION MATRIX:

	Sales	Quantity
Sales	1.000000	0.944922
Quantity	0.944922	1.000000

